(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-104 **Subject: Applied Chemistry**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Do **any six** parts: **(2x6=12)**

(a) All coking coals are caking coals but all caking coals are not coking coals.

(b) Hydrocarbons that are poor gasoline fuels are quite good diesel fuels. Explain

(c) A eutectic mixture has a definete composition and asharpmelting point, yet it is not a compound. Justify.

(d) Describe the method of preparation of the polymer, Nylon-6, 6.

(e) Why is phosphate conditioning better than the carbon aye conditioning?

(f) A water sample contains 248 mg $CaSO_4$ per liter. Calculate the hardness in terms of $CaCO_3$ equivalent.

(g) Why impure metals corrode faster than pure metal under identical conditions?

(h) A pure metal rod half immersed in water starts corroding at the bottom. Give reason.

UNIT-I

Q2 (a) A sample of coal was tested for its calorific value using Bomb's Calorimeter. Following data was obtained.
Weight of coal burnt = 0.920 gm.
Weight of water taken = 550 gm.
Water equivalent of bomb and calorimeter = 2200 gm.
Rise in temperature = 2.42 °C
Fuse wire correction = 10 Cal
Acid Correction = 50 Cal
Assuming coal to contain C = 93%, H = 6% and ash = 1%, Calculate gross and net calorific value of coal if the latent heat of condensation of steam is 580 Cal/gm. **(4)**

(b) What are the advantages of catalytic cracking over thermal cracking? **(4)**

(c) What do you understand by the term knocking in IC engine? Explain the relation between chemical structure and knocking in petrol engine. **(4)**

Q3 (a) A sample of coal was analyzed as follows. Exactly 2.5 g was weighed into a silica crucible. After heating for an hour at 110 °C, the residue weighed 2.415 g. The crucible was then covered with a vented lid and strongly heated for exactly 7 min at 950 ± 20 °C. The residue weighed 1.528 g. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to weigh 0.245 gm. Calculate the percentage of moisture, volatile matter, ash and fixed carbon in the coal sample. **(4)**

(b) What are gaseous fuels? Give some examples. What are the merits and demerits of gaseous fuels over solid and liquid fuels? **(4)**

P.T.O.

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B. tech Second Semester, April, 2024

Paper Code: BS-104 Subject: Applied Chemistry

Time: 1 ½ Hrs. Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	A sample of coal c[illegible]tains C=63%, O=30%, H=6%, S=0.5%, N=0.2% and Ash=0.3%. Calculate the GCV of coal by Dulong's formula.	2	C[illegible] 1
(b)	What is Sweetening of Petrol (with reaction)?	2	CO-1
(c)	With the help of Phase Rule, find the Degree of Freedom of the following systems: (i) Ice & water in equilibrium (ii) Ag & Pb at the Eutectic point (iii) Saturated solution of NaCl (iv) Solid Ag & Soln. of Ag & Pb.	2	CO-2
(d)	Classify the following into Plastics, Elastomers & Fibres: (i)Nylon-6 (ii) HDPE (iii) Natural Rubber (iv) PVC	2	C0-2
(e)	Giv Composition, properties (any 2), Uses (any 2) of Water Gas & Producer Gas.	2	CO-1
Q 2 (a)	Describe how the GCV & NCV of a Solid fuel is determined using a Bomb Calorimeter (with diagram). Also give the formula of GCV with corrections involved.	5	CO-1
(b)	A gaseous fuel has the following composition by volume – CH_4=6%, CO=24%, CO_2=10%, H_2=10%, N_2=50%. If 30% excess air is supplied, calculate the volume of air supplied for 1m^3 of the fuel.	3	CO-1
(c)	Differentiate between Thermal Cracking & Catalytic Cracking.	2	CO-1
Q 3 (a)	With the help of a well labelled diagram explain the Sulphur System. Also explain why in a Sulphur System all the four phases cannot exist in equilibrium.	5	CO-2
(b)	Explain about Eutectic System, Eutectic mixture & Eutectic point.	3	CO-2
(c)	Write about the following polymers (i) Conducting polymers (II) Biodegradable Polymers (iii) LDPE (on any 2).	2	C0-2
Q 4.	Attempt any 2 parts		
(a)	What is the reason for Knocking in IC Engine. Define Octane No. & Cetane No.	5	CO-1
(b)	2.12g of the coal sample was Kjehldahlised and NH_3 gas thus evolved was absorbed in 30ml of 0.1N H_2SO_4. After absorption the excess acid required 10ml of 0.1N NaOH for complete neutralisation. Determine the percentage of Nitrogen.	5	CO-1
(c)	Explain about the Binary System with Congruent melting point & how it is different from Eutectic System.	5	CO-2
(d)	Classify the Polymers on the basis of Tacticity, polymerisation mechanism & on the basis of origin.	5	CO-2

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END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-104 **Subject: Applied Chemistry**

[Batch 2021 Onwards]

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q.1. **Attempt any three.** (3×5=15)

(a) On burning 0.83 g of a solid fuel in a bomb calorimeter, the temperature of 3500 g of water increased from 26.5°C to 29.2°C. Water equivalent of calorimeter and latent heat of steam are 385.0 g and 587.0 cal/g respectively. If the fuel contains 0.7% hydrogen, calculate its gross and net calorific value.

(b) State Gibb's Phase rule. With the help of suitable example explained the terms:
i) Components ii) Degree of freedom iii) Phase

(c) What are conducting polymers? Illustrate with an example.

(d) Calculate the quantity of lime and soda required to soften 20,000 liters of water containing the following salts:
$CaCO_3$ =10.0 mg/L, $MgCO_3$ =8.4 mg/L, $CaCl_2$ =11.1 mg/L, $MgSO_4$ =6.0 mg/L, SiO_2 =1.2 mg/L
Assuming the purity of lime as 90% and soda as 95%.

(e) Discuss sacrificial anodic protection. What is the condition for a metal to act as a sacrificial anode to iron?

UNIT-I

Q.2. (a) Define carbonization of coal.Explain Otto-Hoffman oven method of carbonization and recovery of by-products with diagram. What are its advantages over earlier methods? (10)

(b) What do you understand by the term knocking in IC engine? Explain its significance with chemical constituents of fuels. (5)

OR

Q.3. (a) What are advantages of catalytic cracking of petroleum? Explain fixed bed catalytic method of cracking. Give its mechanism. (10)

(b) Discuss the relative merits and demerits of solid, liquid and gaseous fuels. (5)

UNIT-II

Q.4. (a) What do you mean by congruent melting point? Discuss the phase diagram of a two-component system with the formation of a compound having congruent melting point. (6)

(b) Give the preparation properties and uses of the following (3x3=9)
i) Phenol formaldehyde resin
ii) LDPE
iii) Nylon

P.T.O.